Domain-Specific Search: BM25 vs. Neural Retrieval on Technical Documents

Final Report | Information Retrieval | Group 10

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Repository: https://github.com/surajsinhpatil/bm25-vs-neural

Status: final results on the frozen, adjudicated qrels (1339 judged pairs; checksum and provenance in `data/qrels/`). Per-category results are exploratory (10 queries each). The SciFact numbers (Section 5.1) only validate the implementation and are not cs.CR results. The RQ5 results (Section 5.6) are exploratory: only the default BM25 run was in the judging pool.

1 Introduction

Dense retrievers trained on general web data are often assumed to beat lexical retrieval. We test this on technical text where exact strings matter: computer-security abstracts from arXiv (cs.CR), which are full of identifiers (CVE numbers, protocol versions), homonyms (*oracle*, *poisoning*, *fingerprinting*) and conditions such as exclusions and date limits. We compare our identifier-aware BM25, two dense models used without fine-tuning (E5-base-v2 and SPECTER2), and their Reciprocal Rank Fusion hybrids. Research questions: **RQ1** Does a dense retriever beat BM25 overall? **RQ2** How does this depend on the query type? **RQ3** Does the science-specific SPECTER2 beat the general E5? **RQ4** Does a hybrid beat both of its components? **RQ5** How sensitive is BM25 to k_1, b and tokenisation?

2 Related work

BM25 [1] remains a strong zero-shot baseline: in BEIR [2] it beat many dense models outside their training domain. E5 [3] is trained contrastively on large weakly supervised pair data and then fine-tuned. SPECTER2 [4] is trained on scientific citation data, with task adapters for proximity and ad-hoc search. RRF [5] fuses ranked lists using ranks only. Bruch et al. [6] compare RRF with convex score combination. Dense models are known to handle negation poorly [7]. Our design for significance tests follows Smucker et al. [8]; for assessor disagreement it follows Voorhees [10].

3 Data, queries and systems

Corpus. We use all 39,613 arXiv papers with cs.CR as primary or cross-listed category and a first version in 2016–2025. Withdrawn papers and duplicate titles are removed and LaTeX is cleaned (SHA-256 `fda391ecc5b6...`). A document is its title plus abstract. **Queries.** 50 queries, 10 per category: exact identifier (e.g. "Spectre variant 1 bounds check bypass (CVE-2017-5753)"), homonym ("padding oracle attacks on CBC mode"), negation ("anonymous communication systems that are *not* based on Tor"), time constraint ("attacks on TLS 1.3 published after its 2018 standardisation") and paraphrase ("copying a deployed model by repeatedly querying its prediction interface"). Each query has a description and written relevance criteria. **Systems.** (1) BM25 (k_1=1.2, b=0.75) with an identifier-aware analyzer. It keeps `cve-2021-44228` or `tls1.3` as single tokens and also indexes their parts; Lucene's stopwords are removed (including "not") and the Porter stemmer is applied. (2) E5-base-v2 and (3) SPECTER2 (proximity adapter for documents, ad-hoc query adapter for queries), both with exact cosine search. (4, 5) RRF (k=60, top 100 of each run) of BM25 with E5 and with SPECTER2. Nothing was tuned on our queries.

4 Evaluation protocol

Pooling and judging. The top 10 of all five runs form the pool: 1339 query–document pairs, about 27 per query. Every query was judged independently by two of the four authors, with the six annotator pairs spread evenly. The judging pages showed no system names or ranks, and showed dates only for time queries. Grades: 0 not relevant, 1 on topic but does not answer the query or breaks a condition, 2 relevant, 3 highly relevant. All 2,678 judgments were collected and audited: none is missing, duplicated or outside the pool. **Agreement.** Weighted Cohen's $\kappa = 0.760$ (linear) and 0.829 (quadratic); κ on binary relevance (grade ≥ 2) = 0.795; exact agreement 0.757. This meets the protocol's target of $\kappa \geq 0.6$. 325 pairs conflict: 21 are 0 vs 3, and 134 fall on opposite sides of the relevance line. **Adjudication.** All conflicts were resolved in a team discussion and recorded by one author; every final grade is one of the two submitted grades. There was no independent adjudicator: the annotators of a pair took part in its discussion. The recorder was one of the two annotators in 160 conflicts and kept his own grade in 41% of them (51–55% for the other three). 27 decisions were taken from the title or by a "lower grade when unsure" rule, as their rationales record. The qrels were then frozen with a SHA-256 checksum and a provenance file. **Metrics.** The primary metric is nDCG@10 with gain equal to the grade. P@5, MRR@10 and MAP@10 use grade ≥ 2. Unjudged documents count as non-relevant; Hole@10 reports how often they occur. Our metric

Table 1: Effectiveness on the 50 cs.CR queries (frozen, adjudicated qrels; nDCG@10 with 95% bootstrap CI). Binary metrics use grade ≥ 2; Hole@10 = share of top-10 documents never judged.

System	nDCG@10	nDCG@5	P@5	MRR@10	MAP@10	Hole@10
BM25	0.689 [0.63, 0.75]	0.667	0.516	0.632	0.325	0.0%
E5	0.713 [0.65, 0.77]	0.693	0.572	0.693	0.366	0.0%
SPECTER2	0.543 [0.47, 0.61]	0.542	0.400	0.596	0.210	0.0%
BM25+E5	0.744 [0.69, 0.80]	0.717	0.560	0.700	0.369	0.0%
BM25+SPECTER2	0.681 [0.62, 0.74]	0.675	0.524	0.658	0.311	0.0%

Table 2: Paired randomisation tests on nDCG@10 over the 50 queries (100,000 permutations; Holm–Bonferroni over all 10 pairs; $\alpha = 0.05$).

Comparison	ΔnDCG@10	p_{Holm}	Significant
BM25 − E5	−0.023 [−0.084, +0.038]	0.929	no
BM25 − SPECTER2	+0.147 [+0.070, +0.228]	0.005	yes
BM25 − BM25+E5	−0.055 [−0.092, −0.019]	0.023	yes
BM25 − BM25+SPECTER2	+0.009 [−0.034, +0.056]	0.929	no
E5 − SPECTER2	+0.170 [+0.104, +0.241]	<0.001	yes
E5 − BM25+E5	−0.031 [−0.072, +0.010]	0.592	no
E5 − BM25+SPECTER2	+0.032 [−0.026, +0.091]	0.901	no
SPECTER2 − BM25+E5	−0.201 [−0.276, −0.130]	<0.001	yes
SPECTER2 − BM25+SPECTER2	−0.138 [−0.193, −0.087]	<0.001	yes
BM25+E5 − BM25+SPECTER2	+0.063 [+0.021, +0.111]	0.031	yes

code matches `trec_eval` (via `pytrec_eval`) on every query of every run, with a maximum difference of 3×10^{-16}. **Significance.** Paired randomisation tests over queries (100,000 permutations) on nDCG@10 for all 10 system pairs, with Holm–Bonferroni correction. Per-category results come with 95% bootstrap CIs and are exploratory, since each category has only 10 queries.

5 Results

5.1 Validation of the implementation (SciFact)

This is not a cs.CR result. On BEIR SciFact (300 queries) with the same settings as Pyserini, our BM25 differs from Lucene by ΔnDCG@10 = -0.0009 and shares 98.0% of its top-10 documents. On the same collection E5 (0.727) beats BM25 (0.677) significantly (p_{Holm} = 0.023), while SPECTER2 (0.646) does not.

5.2 Main results on cs.CR

Hole@10 is 0 for every system: every document in every top 10 was judged (Table 1). **RQ1:** E5 and BM25 are statistically indistinguishable (E5 +0.023, 95% CI −0.038 to +0.084, $p_{\text{Holm}} = 0.93$). E5 is better on 24 queries and BM25 on 23. The data give no support for the claim that a general dense model beats BM25 on this collection. **RQ3:** SPECTER2 is the weakest system, 0.147 below BM25 and 0.170 below E5 (both $p_{\text{Holm}} \leq 0.005$), and significantly below both hybrids. Its citation-based training does not transfer to short ad-hoc queries. **RQ4:** BM25+E5 is the best system (nDCG@10 0.744). It is significantly better than BM25 (+0.055, $p_{\text{Holm}} = 0.023$; better on 32 queries, worse on 16) and than BM25+SPECTER2 (+0.063, $p_{\text{Holm}} = 0.031$), but not than E5 (+0.031, $p_{\text{Holm}} = 0.59$). So the hybrid does not beat *both* of its components. BM25+SPECTER2 recovers most of SPECTER2's loss but does not beat BM25. Excluding the query with no relevant document leaves every significance decision unchanged. **How much depends on adjudication** (Table 3, computed from the two annotators' grades alone). No way of resolving the conflicts can reverse the four SPECTER2 deficits, and they are significant in 99–100% of 500 random adjudications. BM25+E5 has the highest mean in 100% of them. By contrast, its significant advantage over BM25 appears in only 7% of random adjudications, and the sign of that difference could even flip under some adjudications. That significance is therefore a property of our adjudication rather than a robust effect, and we treat it with caution.

5.3 By query type (RQ2, exploratory)

BM25 leads on *exact identifiers* (0.798 vs. 0.662 for E5), while E5 leads on *negation* (0.677 vs. 0.560) and *paraphrases* (0.692 vs. 0.585; Table 4). BM25+E5 is best on time (0.752) and paraphrase (0.773) queries. On homonyms, BM25, E5 and BM25+E5 are all near 0.84. With 10 queries per category the CIs overlap widely (Fig. 1), so these are hypotheses

Table 3: Sensitivity to adjudication (computed from the two annotators' grades, independent of the adjudicated grades). Exact range of each mean difference over every adjudication that picks one of the two submitted grades (40,269 per-query combinations enumerated); share of 500 random adjudications in which the pair is significant (Holm, 10 pairs); and the test on the single most adversarial adjudication.

Comparison	Exact range of ΔnDCG@10	Sign can flip	Sig. in random adj.	Worst case
BM25 − E5	−0.099 to +0.068	yes	0%	–
BM25 − SPECTER2	+0.001 to +0.250	no	99%	n.s.
BM25 − BM25+E5	−0.092 to +0.015	yes	7%	–
BM25 − BM25+SPECTER2	−0.064 to +0.088	yes	0%	–
E5 − SPECTER2	+0.020 to +0.262	no	100%	n.s.
E5 − BM25+E5	−0.080 to +0.035	yes	0%	–
E5 − BM25+SPECTER2	−0.067 to +0.118	yes	0%	–
SPECTER2 − BM25+E5	−0.283 to −0.046	no	100%	n.s.
SPECTER2 − BM25+SPECTER2	−0.202 to −0.032	no	100%	n.s.
BM25+E5 − BM25+SPECTER2	−0.021 to +0.122	yes	16%	–

Table 4: nDCG@10 by query category (10 queries each; exploratory). 95% CIs in Fig. 1.

System	Exact identifier	Homonym	Negation	Time constraint	Paraphrase
BM25	0.798	0.839	0.560	0.665	0.585
E5	0.662	0.846	0.677	0.688	0.692
SPECTER2	0.469	0.577	0.612	0.597	0.460
BM25+E5	0.755	0.845	0.596	0.752	0.773
BM25+SPECTER2	0.699	0.813	0.592	0.678	0.622

rather than findings. Within-category tests (Holm within each category) find SPECTER2 below each of the other four systems on homonyms, and BM25+E5 above BM25 on paraphrases.

5.4 Error analysis

Conditions are ignored. On negation queries, 81% of BM25's top 10 are graded 1, mostly papers that break the exclusion. The figure is 60% for E5 and 71% for BM25+E5 (Table 5). BM25 removes "not" as a stopword: for "anonymous communication systems that are not based on Tor", seven of its top 10 titles name Tor (e.g. "Analyzing Trends in Tor", "DarkHorse: … Tor Onion Services"), all graded 1. E5's top 10 includes DC-nets, Atom and Pirates (grade 3). On time queries, 40–54% of each system's top 10 is graded 1. Counting from first-version dates, 30–38% of each system's top 10 breaks the date condition; a rule-based date filter (stretch goal S2) removes all of these. **Paraphrases defeat lexical matching.** For "programs that lock a victim's files and demand payment to unlock them", BM25 ranks "Scalable Time-Lock Puzzle" and a digital-payment framework near the top (nDCG@10 0.18), while E5 returns ransomware papers (0.82). **Identifiers defeat dense matching.** For "replay attacks on TLS 1.3 0-RTT early data", E5's top five are replay attacks in unrelated fields such as space launch and sharded ledgers (0.18), while BM25 finds TLS 1.3 papers (0.82). **The two systems complement each other.** Of the relevant documents in the BM25 and E5 top 10s, 110 are found by both, 125 only by BM25 and 154 only by E5. This is why fusion helps. Full examples are in `deliverables/final/analysis/error_analysis.md`.

5.5 Efficiency

All numbers are measured from the run logs (Table 6). BM25 builds its index in 20 s on one CPU into 29 MB and answers in 1.6 ms. Each dense index is 116 MB (4× larger) and takes about 3 min of T4 GPU time. Dense queries take 17–30 ms, mostly query encoding; exact search alone takes 6–7 ms. BM25+E5 therefore buys its effectiveness at roughly 12× the query cost of BM25.

5.6 BM25 sensitivity (RQ5, exploratory)

Both experiments use the corpus, the 50 queries, the frozen qrels and the metric code of the main experiment. They ran on Kaggle (commit `df51119`) after three checks: the qrels match their frozen checksum, the rebuilt corpus is the frozen one (39,613 documents, SHA-256 `fda391ecc5b6...`), and the default BM25 run rebuilt from it reproduces the pooled run (same top 10 for all 50 queries, scores within 4×10^{-6}). Outputs, metadata and checksums are in `deliverables/rq5/`.

Pool bias. Only one BM25 configuration, the default (k_1=1.2, b=0.75, identifier-aware analyzer), was in the judging pool. Every other setting puts documents nobody judged into its top 10, and nDCG@10 counts them as non-relevant, which favours the default by construction. We therefore also report Hole@10 (the share of unjudged documents in the top 10)

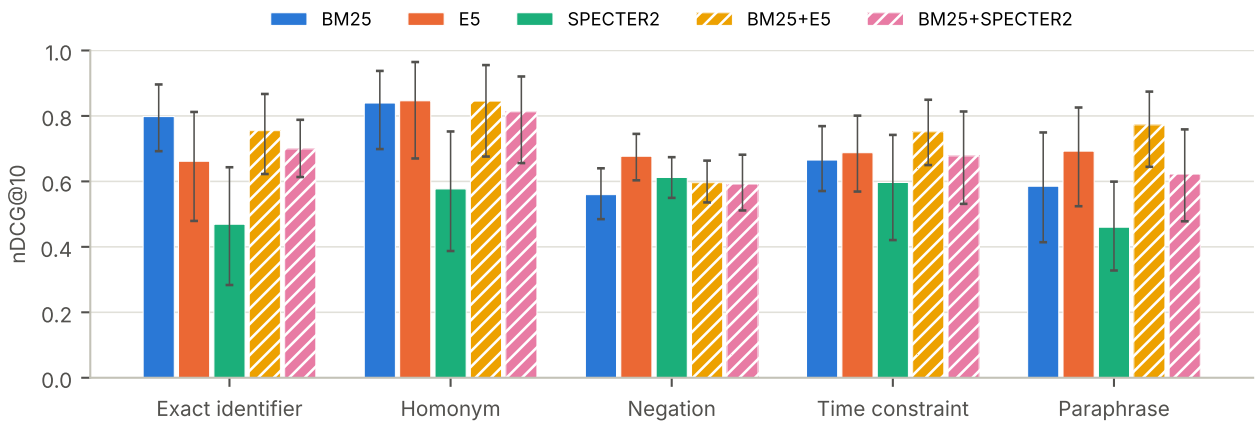


Figure 1: nDCG@10 per category with 95% bootstrap CIs (hatched = hybrids). Exploratory.

Table 5: Share of each system's top 10 graded 1 (on topic but does not answer, or breaks the exclusion / date limit).

System	Exact identifier	Homonym	Negation	Time constraint	Paraphrase
BM25	0.38	0.06	0.81	0.54	0.14
E5	0.31	0.05	0.60	0.46	0.15
SPECTER2	0.31	0.11	0.69	0.40	0.25
BM25+E5	0.37	0.05	0.71	0.48	0.14
BM25+SPECTER2	0.41	0.07	0.74	0.45	0.15

and condensed nDCG@10 [9], which removes unjudged documents from the ranking before cutting at 10. Condensed nDCG@10 is not a bound or an unbiased estimate either: it can overrate runs that were not pooled, and in our data it rises with Hole@10. We treat a result as robust only if both metrics agree. Across the 132 settings, Hole@10 correlates with nDCG@10 at r=−0.97 and with condensed nDCG@10 at r=0.86 (Fig. 2). Away from the default, both metrics mostly reflect how many unjudged documents a setting retrieves.

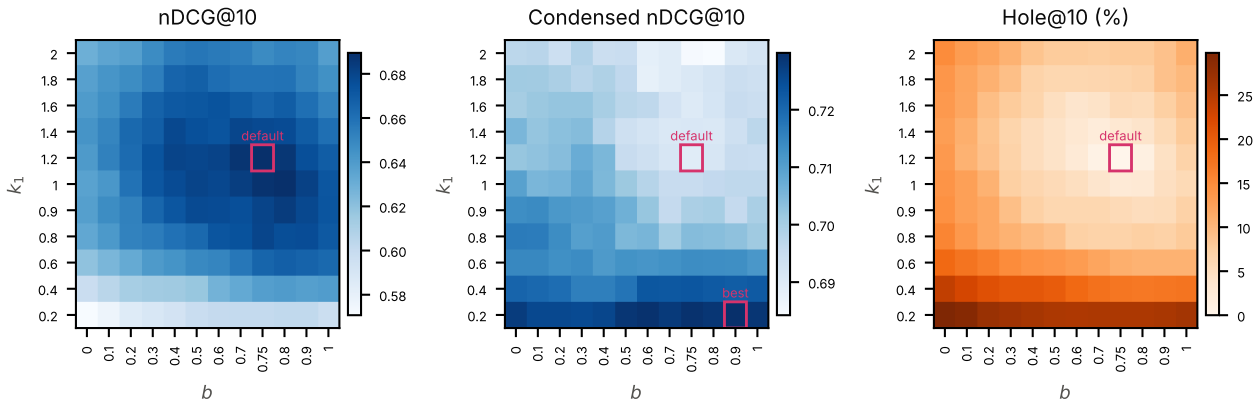


Figure 2: RQ5a (exploratory): nDCG@10, condensed nDCG@10 and Hole@10 over the (k_1, b) grid. Boxes: the default setting and the best setting on condensed nDCG@10. The two metrics favour opposite regions, and condensed nDCG@10 is highest where most of the top 10 was never judged.

k_1 **and** b (Table 7). On nDCG@10, 0 of the 132 settings score above the default. 12 settings lie within 0.01 of it (k_1 0.9–1.2, b 0.4–0.9), and 32% lie within 0.02. Low k_1 looks clearly worse ($k_1 \leq 0.4$: 0.571–0.640), but these are the settings with the most unjudged documents. Condensed nDCG@10 orders the grid almost the other way round. Its best setting is k_1=0.2, b=0.9 (0.730), where 25.8% of the top 10 is unjudged and nDCG@10 is 0.602. Neither ranking of settings far from the default can be trusted without judging their documents. The default being the best setting on nDCG@10 is what pool bias predicts, so it is not evidence that the default is optimal. The only tuned estimate is cross-validated: 5 folds stratified by category, with (k_1, b) chosen on four folds and scored on the fifth. It reaches 0.684 against 0.689 for the default (Δ = −0.006, 95% CI −0.011 to −0.002, p=0.012; better on 8 queries, equal on 30, worse on 12). On condensed nDCG@10 the difference vanishes (−0.0001, p=0.957). Because the metrics disagree, we do not claim that tuning hurts; we only conclude that it did not help. The folds chose (1, 0.8), (1, 0.8), (1.2, 0.75), (1, 0.8), (1.2, 0.75), and 20 other fold assignments give 0.679–0.687.

Tokenisation (Table 8). *Identifier handling* made no measurable difference. Replacing the identifier-aware analyzer

Table 6: Efficiency on the cs.CR corpus. Dense indexing = encoding the corpus on the GPU; dense query time includes encoding the query; hybrid = sum of its components (p95 not defined).

System	Index (s)	Index (MB)	Query (ms)	p95 (ms)	Hardware
BM25	20.3	29.3	1.6	1.7	cpu
E5	174.6	116.0	17.3	18.2	Tesla T4
SPECTER2	182.6	116.0	30.0	36.6	Tesla T4
BM25+E5	194.9	145.4	19.3	–	–
BM25+SPECTER2	202.8	145.4	32.0	–	–

Table 7: RQ5a (exploratory): BM25 (identifier-aware analyzer) over 132 (k_1, b) settings on the 50 cs.CR queries. $\Delta\,(p)$: cross-validated minus default, paired randomisation test. Only the default run was in the judging pool; nDCG@10 counts unjudged documents as non-relevant, condensed nDCG@10 removes them, Hole@10 is the share of unjudged documents in the top 10. †Chosen on the same queries it is scored on: sensitivity only, not a performance estimate.

BM25 setting	nDCG@10	$\Delta\,(p)$	Cond. nDCG@10	$\Delta\,(p)$	Hole@10
Default (k_1=1.2, b=0.75)	0.689	–	0.689	–	0.0%
5-fold CV, seed 11	0.684	−0.006 (0.012)	0.689	−0.0001 (0.957)	1.8%
CV, 20 other fold seeds	0.679–0.687				
Anserini default (k_1=0.9, b=0.4)	0.672		0.710		7.2%
Best, nDCG@10† (k_1=1.2, b=0.75)	0.689		0.689		0.0%
Best, cond.† (k_1=0.2, b=0.9)	0.602		0.730		25.8%
Range, all 132 settings	0.571–0.689		0.684–0.730		0%–30%

with the Lucene-like one changes nDCG@10 by −0.003 (p_{Holm}=0.267), 41 of 50 queries are unchanged, and condensed nDCG@10 agrees. This is the one comparison almost free of pool bias: Hole@10 is 0.8% (4 unjudged documents in total). Even on the identifier queries both analyzers score 0.798. A plausible reason, which we did not test, is that these queries also contain distinctive words ("Log4Shell", "Spectre"), and the Lucene-like analyzer still indexes the pieces of each identifier. *Stemming* cannot be judged with the current qrels. Removing it lowers nDCG@10 by −0.089 (p_{Holm} <0.001), but raises Hole@10 to 23.6%, and condensed nDCG@10 moves the other way (+0.033, p_{Holm}=0.021). Both differences are significant and in opposite directions, so we draw no conclusion; the same holds without the identifier handling. *Keeping stopwords* (including "not") gives −0.016 on nDCG@10 (p_{Holm}=0.096) and +0.014 on condensed nDCG@10 (p_{Holm}=0.069), so no conclusion either. On the negation queries it scores 0.526 against 0.560, so keeping "not" as a term shows no benefit for negation here (with 6.0% of its top 10 unjudged overall). Judging the 141 unjudged query–document pairs in the variants' top 10s would resolve these comparisons.

Answer to RQ5. Near the default, BM25 is insensitive to k_1 and b: there is a broad plateau, and cross-validated tuning gave no gain. Identifier-aware tokenisation made no measurable difference on our queries. Whether stemming, stopword removal or settings far from the default matter cannot be decided with the current judgments.

Table 8: RQ5b (exploratory): tokenisation ablation, k_1=1.2, b=0.75 for every analyzer; all use Lucene's stopwords and the Porter stemmer unless stated. $\Delta\,(p_{\text{Holm}})$: difference from the main analyzer, paired randomisation test, Holm over the 4 comparisons, for nDCG@10 and for condensed nDCG@10 separately. W/T/L: queries where the variant is better / equal / worse on nDCG@10. Only the main run was in the judging pool, so a high Hole@10 penalises a variant in nDCG@10 for documents nobody judged.

Analyzer	nDCG@10	$\Delta\,(p_{\text{Holm}})$	Cond. nDCG@10	$\Delta\,(p_{\text{Holm}})$	Hole@10	W/T/L
Identifier-aware (main)	0.689	–	0.689	–	0.0%	–
Lucene-like	0.687	−0.003 (0.267)	0.690	+0.001 (0.988)	0.8%	3/41/6
Identifier-aware, no stem.	0.600	−0.089 (<0.001)	0.722	+0.033 (0.021)	23.6%	13/4/33
Lucene-like, no stem.	0.599	−0.090 (<0.001)	0.722	+0.032 (0.021)	23.8%	12/3/35
Identifier-aware, stopwords kept	0.673	−0.016 (0.096)	0.704	+0.014 (0.069)	6.0%	18/9/23

6 Limitations

(1) There are 50 queries, 10 per category, so category effects are exploratory. (2) The pool contains only the top 10 of five systems. Hole@10 is 0 for these systems, but recall-oriented metrics, and any system outside the pool, would be biased. (3) Adjudication was a team discussion without an independent adjudicator, and 27 decisions rest on titles or a "lower when unsure" rule. The hybrid's significant advantage over BM25 is sensitive to such choices (Table 3). (4) No relevant document (grade ≥ 2) was found for one query (AES-GCM nonce reuse); excluding it changes no conclusion. (5) The authors wrote the queries and judged them. (6) The dense models may have seen arXiv abstracts during pre-training. (7) A grade of 1 earns nDCG gain, so systems partly get credit for documents that break a condition. With an exploratory strict gain (no gain for grade 1, `sNDCG@10`), the order is the same except that BM25+SPECTER2 (0.533) moves just above BM25 (0.523). (8) RQ5 is exploratory: only the default BM25 configuration was pooled, so other settings and analyzers are scored with unjudged documents in their top 10 (up to 30%), and ordinary and condensed nDCG@10 disagree for most of them. The cross-validation reuses the same 50 queries.

7 Conclusions

On security abstracts, a general dense retriever used without fine-tuning does not beat a careful BM25 overall, and the science-specific SPECTER2 is clearly worse than both. BM25 and E5 fail on different queries: BM25 on paraphrases and negation, E5 on exact identifiers. As a result, their RRF fusion is the best system (nDCG@10 0.744), significantly better than BM25, although that margin depends on how the judgment conflicts were adjudicated. It is not better than E5 alone, and it costs about $12\times$ the query time of BM25. Every system largely ignores exclusions and date limits; handling them explicitly (S2) looks more promising than switching models. BM25 itself is robust near its default parameters, and identifier-aware tokenisation made no measurable difference. Whether stemming or stopword removal matter remains open until the documents those variants retrieve are judged.

Contribution matrix

Member	Components
Surajsinh Patil	[to be filled in by the group]
Tejas Mahajan	[to be filled in by the group]
Swayam Kumud	[to be filled in by the group]
Saksham Singhal	[to be filled in by the group]
All	Relevance judging (635–697 judgments each, 2,678 in total); resolving the 325 judgment conflicts in a team discussion (recorded by Surajsinh Patil)

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